

CAPSTONE PROJECT PROPOSAL

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Adaptive CCTV Compression System Using Dynamic Region of Interest (ROI) Control

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in partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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DECLARATION

We hereby declare that the work presented in this Capstone Project Proposal is the result of our own investigation and efforts, conducted under the guidance and supervision of Hasan Abdullah, Assistant Professor, Department of Computer Science and Engineering, Bangladesh Army International University of Science and Technology, Cumilla, Bangladesh. We confirm that no part of this proposal has been submitted, nor is being submitted, for the award of any other degree or diploma at any institution.

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1. Introduction

Surveillance systems are fundamental to security infrastructure in smart cities, educational institutions, industrial facilities, and public spaces across Bangladesh. The rapid expansion of urban populations and the government's Smart Bangladesh Vision 2041 have driven an unprecedented deployment of CCTV cameras nationwide. However, conventional surveillance systems transmit video at fixed high bitrates, resulting in excessive bandwidth consumption, prohibitive storage costs, high latency, and degraded performance in resource-constrained network environments [1], [2].

This project presents an Adaptive CCTV Compression System with Dynamic Region of Interest (ROI) Control. The system employs YOLOv8-based intelligent object detection on an edge computing node to identify and classify regions of interest in real time, combined with adaptive H.265/HEVC compression via FFmpeg. Static background areas are aggressively compressed while moving and semantically important regions (persons, vehicles) are preserved at high visual quality. A central server with an interactive web dashboard provides users with real-time control over compression parameters and live visualization of system performance metrics including bitrate, PSNR, SSIM, and VMAF scores [3], [4].

Beyond the originally proposed scope, the implemented system incorporates a risk-aware event engine with a three-state surveillance state machine (NORMAL, ALERT, CRITICAL) driven by a multi-signal risk score, temporal ROI persistence with IoU-based smoothing to eliminate visual flicker, multi-codec runtime switching (H.264, H.265, AV1, NVENC), privacy-preserving operation modes (background blur, ethical blanking, face masking), and a comprehensive evaluation pipeline that computes per-ROI and background-only quality metrics alongside detection accuracy retention and VMAF perceptual quality scoring [5], [6].

2. Research Motivation

Bangladesh's CCTV surveillance infrastructure has experienced explosive growth over the past decade. The national CCTV market has grown at approximately 50% per year, with camera installations rising from 200,000 units in 2015 to over 1.6 million units by 2024. The smart surveillance segment is projected to expand from USD 88.9 million in 2025 to USD 148.7 million by 2030, driven by rapid urbanization, mega-infrastructure projects, metro rail development, and the Smart Bangladesh Vision 2041 initiative aiming for IoT-enabled smart cities and comprehensive digital security [7], [8].

Despite this growth, conventional CCTV systems rely on fixed high-bitrate streaming protocols that lead to severe inefficiencies: 40-50% of available bandwidth is wasted on static background content in typical surveillance deployments. This places enormous strain on Bangladesh's internet infrastructure, which recorded 37% peak traffic growth in 2025 alone. Educational institutions, industrial facilities, and public safety systems bear significant operational costs for storage and bandwidth with no corresponding security benefit [9], [10].

Key motivations behind this research are:

- Reduce bandwidth consumption by 45-65% while preserving high visual quality in critical moving regions through dynamic ROI control based on YOLOv8 object detection with priority classification.
- Implement edge-based real-time adaptive compression with H.265/HEVC streaming suitable for resource-constrained networks and low-cost IoT deployments in Bangladesh.
- Provide efficient and scalable surveillance technology in support of the government's Digital Bangladesh and Smart City initiatives.

- Reduce storage and energy costs for educational institutions, industries, and public safety systems through aggressive background compression during idle periods.
- Bridge the gap between traditional fixed-bitrate CCTV systems and modern intelligent surveillance by introducing a risk-aware event engine that preserves evidence quality during critical incidents.
- Develop a sustainable and cost-effective smart surveillance infrastructure aligned with the national development goals of Bangladesh.

3. Problem Statement

The rapid proliferation of CCTV surveillance systems in Bangladesh has introduced significant technical and economic challenges. With over 1.6 million cameras installed nationwide and the smart surveillance market expanding at 50% annually, conventional systems continue to stream video at fixed high bitrates, leading to critical inefficiencies [7], [11].

Existing CCTV systems suffer from the following major disadvantages:

- **High Bandwidth and Storage Waste:** Fixed-bitrate streaming wastes 40-50% of available network capacity even when the vast majority of the scene remains static.
- **No Intelligent Adaptation:** Conventional systems lack dynamic adjustment of video quality based on motion, semantic importance, or threat level, resulting in unnecessary data transmission.
- **Poor Performance on Resource-Constrained Networks:** Bangladesh's low-bandwidth IoT environments frequently experience high latency, packet loss, and degraded video quality.
- **No Real-Time User Control:** There is no live dashboard for operators to dynamically adjust compression parameters based on situational needs or threat levels.
- **Inefficient Resource Utilization:** Educational institutions, industries, and public safety systems expend significant operational costs with no proportional security benefit.
- **Lack of Event-Aware Preservation:** Existing compression methods do not distinguish between routine idle monitoring and critical security incidents, potentially destroying forensic evidence through aggressive compression during events.

These challenges impede the successful deployment of smart surveillance and IoT-based security solutions in Bangladesh. There is an urgent need for an adaptive, edge-based compression system that intelligently preserves quality in semantically important regions while significantly reducing bandwidth consumption, and that automatically transitions to evidence-preservation mode during security incidents.

4. Main Objectives

4.1 Specific Goals

4.1.1 Intelligent ROI Detection and Compression

Develop an edge node that combines YOLOv8 neural object detection with priority-based classification (HIGH for persons, MEDIUM for vehicles, LOW for other objects) and FFmpeg adaptive compression with runtime codec switching (H.264, H.265, AV1, NVENC). Moving regions are preserved at high quality while static background is aggressively compressed.

4.1.2 Risk-Aware State Machine and Event Engine

Implement a multi-signal risk score function that combines object presence, motion area, time-of-day, scene change magnitude, and ROI density to drive a three-state surveillance machine (NORMAL, ALERT, CRITICAL) with hysteresis. A pre/post-event frame buffer preserves up to 15 seconds of forensic context before and after critical events.

4.1.3 Server-Side Stream Management and Metrics Collection

Build a central server that manages dual Socket.IO namespaces for stream relay, monitors UDP H.265 bitrate in real time, performs automatic bandwidth adaptation, logs comprehensive metrics (bitrate, latency, FPS, PSNR, SSIM, risk scores) to CSV, and provides REST APIs for evaluation and playback.

4.1.4 Interactive Live Dashboard

Create a real-time web-based dashboard (Next.js) with canvas-based live stream rendering, priority-coded ROI overlay with glow effects, interactive control sliders (BG Scale, BG Quality, ROI Quality, Detect Every N Frames), preset modes, dual bandwidth display, live Chart.js bandwidth visualization, and real-time state badge showing NORMAL/ALERT/CRITICAL status.

4.1.5 Comprehensive Performance Evaluation

Develop an automated evaluation pipeline that computes global PSNR/SSIM, per-ROI PSNR/SSIM, background-only PSNR/SSIM, VMAF perceptual quality scores, detection accuracy retention, and an investigation usability score. Generate visual artifacts including diff heatmaps and summary plots.

4.1.6 Scalable Deployment

Containerize the full stack with Docker and Docker Compose for reproducible deployment. Support YAML configuration profiles for balanced, high-quality, privacy, and ultra-low-bandwidth operation modes.

5. Research Questions

- How effectively can dynamic ROI-based compression with YOLOv8 object detection reduce bandwidth consumption compared to traditional fixed-bitrate CCTV systems?
- How well does the suggested system compress background areas while preserving visual quality (as determined by PSNR, SSIM, and VMAF) in semantically important regions?
- What effects does YOLOv8-based real-time object detection have on the precision and effectiveness of ROI selection in surveillance video streams compared to conventional motion detection?
- How do user-controlled compression parameters (background quality, ROI quality, detection interval) affect bandwidth consumption and system performance under varying network conditions?
- How effectively does the risk-aware event engine preserve forensic evidence quality during critical incidents compared to uniform compression approaches?
- How well-suited is the proposed edge-server architecture with Docker containerization for deployment in IoT contexts with limited resources, such as those in Bangladesh?

6. Literature Review

Recent advances in Region of Interest (ROI)-based video compression have demonstrated significant potential for bandwidth efficiency in surveillance systems. This section reviews contemporary approaches and identifies the research gap addressed by this project.

6.1 Literature Review Summary

SL	Year	Methodology	Dataset	Review / Gap
1	2025	FFmpeg metadata (motion vectors, bitrate maps) and morphological operations for ROI; adaptive YOLOv8 model scheduling [12]	YODA dataset (highway traffic videos, 1280x720@30fps)	Strong latency reduction; Gap: Best for static scenes only; no dashboard or risk-aware compression
2	2024	CNN-based lambda prediction reservoir model and HEVC R-lambda QP adjustment at frame/CU level [13]	HEVC test sequences (Class B-F: sports, news, movies)	Excellent ROI quality with low bitrate overhead; Gap: Needs high-quality ROI maps; limited real edge testing
3	2023	Gaussian Mixture Model (GMM) ROI detection, HEVC All-Intra tiling, cross-layer packet prioritization [14]	Highway overtaking video (simulation in SUMO+NS-2)	Good ROI PSNR in lossy networks; Gap: Simulation only; weak on complex indoor CCTV scenes
4	2023	LightweightROIDet (YOLOv5-Lite, block motion), dynamic bandwidth allocation, FFmpeg [15]	Real multi-camera surveillance streams	54% bandwidth saving with minimal accuracy drop; Gap: Only for AI tasks; no human-viewing PSNR/SSIM or dashboard
5	2024	Dual-scale compressive sensing, Siamese network ROI extraction, reversible NN reconstruction [16]	Private transmission-line surveillance and CD.Net (with noise)	High PSNR/SSIM at very low bitrate; Gap: High parameter sensitivity and reconstruction overhead
6	2023	Fast pre-processing ROI detection, MJPEG with ROI-aware bitrate allocation [17]	Low-power wireless surveillance sequences	Energy and bitrate reduction; Gap: Uses outdated MJPEG; no dashboard or multi-camera support
7	2025	Macroblock importance predictor (MobileSeg), region-aware super-resolution enhancer [18]	YODA, YouTube clips, BDD100K	2-3x faster throughput and higher accuracy; Gap: Requires retraining per task; no dashboard
8	2025	YOLOv9 for frame relevance, SSIM similarity check, FFmpeg/OpenCV compression [19]	Real ATM bank camera videos (15 hours, day/night)	85% average compression with 100% important frames kept; Gap: Storage-focused, not real-time streaming or risk-aware
9	2025	Reinforcement Learning (PPO) agent for block-level QP control in x264 [20]	BDD100K driving videos	25% better BD-rate for detection tasks; Gap: Needs downstream AI task model; machine-vision only
10	2025	Real-time ROI detection and adaptive encoding with intelligent frame caching [21]	Real mega-pixel CCTV streams	Significant bandwidth saving with high ROI quality; Gap: No public dashboard or risk-aware event engine
11	2025	ROI detection, frame retargeting (stretch ROI, shrink background), standard codec [22]	Standard VCM/machine-vision test sequences	Better machine-task accuracy at same bitrate; Gap: Extra retargeting step; mainly for machines
12	2024	Transformer-based video compression with learned ROI weighting [23]	UVG, MCL-JCV datasets	State-of-the-art compression ratios; Gap: High computational cost, unsuitable for real-time edge deployment

SL	Year	Methodology	Dataset	Review / Gap
13	2024	Adaptive bitrate streaming with SDN control for surveillance networks [24]	Custom multi-camera testbed	Dynamic bandwidth allocation across cameras; Gap: No content-aware ROI encoding or event detection
14	2023	Privacy-preserving video analytics with selective encryption and ROI masking [25]	VIRAT, PRID-2011 datasets	Strong privacy guarantees with low overhead; Gap: Focused on privacy, not compression efficiency or bandwidth reduction
15	2025	Federated learning for distributed surveillance video analysis [26]	Custom distributed camera network	Privacy-preserving distributed inference; Gap: High communication overhead; not optimized for compression

6.2 Comparative Analysis

Wang and Yang [12] proposed a compression metadata-assisted ROI extraction framework for H.265/HEVC coding focused on edge-based video analytics, achieving latency reduction and improved F1-score for detection tasks. However, their approach lacks perceptual quality metrics (PSNR, SSIM) and any form of user-interactive monitoring or visualization. Dou et al. [13] presented an R-lambda optimization based on CNN with ROI-based coding in the HEVC framework, achieving better perceptual quality in ROI regions with low bitrate overhead. However, implementation is limited to simulation environments without real-world edge deployment validation.

Labiod et al. [14] proposed GMM-based ROI detection with cross-layer optimization for VANETs, claiming up to 11 dB ROI PSNR improvement, but evaluation was simulation-only without actual bandwidth reduction measurements. Guo et al. [15] developed a bandwidth-efficient streaming architecture using YOLOv5-Lite with approximately 54% bandwidth reduction, but evaluation focused on analytics accuracy rather than human-perceived visual quality. Gao et al. [16] proposed a robust mixed-rate ROI-aware compressive sensing framework achieving PSNR of 34.71 dB and SSIM of 0.932, but real-time surveillance application is constrained by computational overhead of signal reconstruction.

More recent approaches have leveraged deep learning for enhanced compression. Agrawal et al. [19] proposed FRVC using YOLOv9 for frame relevance detection, achieving up to 85% storage reduction, but the system is optimized for storage rather than real-time streaming. Gadot et al. [20] proposed reinforcement learning-based rate control achieving 25% BD-rate reduction through block-level optimization, but this is task-specific and designed for machine vision rather than human perceptual quality. Transformer-based approaches [23] offer state-of-the-art compression but remain too computationally expensive for real-time edge deployment.

Chuang et al. [21] proposed adaptive encoding and caching schemes for surveillance streaming, and Rozek et al. [22] developed ROI-based retargeting techniques for machine-oriented video coding. SDN-based adaptive streaming approaches [24] provide dynamic bandwidth allocation across multiple cameras but lack content-aware ROI encoding. Privacy-preserving techniques [25] and federated learning approaches [26] address important adjacent concerns but do not directly target compression efficiency or event-aware quality preservation.

6.3 Research Gap and Proposed Contribution

The literature analysis reveals that existing approaches are predominantly focused on analytics performance, simulation-based validation, or machine-centric optimization. Critically, no existing work simultaneously addresses: (a) real-time edge deployment with neural object detection, (b)

user-controlled adaptive compression with live visualization, (c) risk-aware event-driven quality preservation, and (d) comprehensive evaluation spanning signal quality, perceptual quality (VMAF), and investigation usability metrics.

This research presents a complete end-to-end adaptive CCTV compression system that addresses all four dimensions. The framework integrates YOLOv8-based object detection at the edge node, FFmpeg adaptive encoding with runtime codec switching, a multi-signal risk score driving a three-state surveillance machine, pre/post-event frame buffering for forensic evidence preservation, and an interactive dashboard enabling real-time user control. The evaluation pipeline computes global, per-ROI, and background-only PSNR/SSIM alongside VMAF scores, detection accuracy retention, and a novel investigation usability composite score. Experimental results demonstrate bandwidth reduction of 45-65% with stable perceptual quality and enhanced evidence preservation during critical incidents.

7. Project Scope

The goal of this project is to design, build, and evaluate a complete Adaptive CCTV Compression System that leverages Dynamic Region of Interest (ROI) Control with a risk-aware event engine. The project delivers a fully functional prototype demonstrating intelligent surveillance with real-time adaptive video compression.

7.1 Within Scope

- Development of an edge node with YOLOv8-based object detection that identifies and classifies ROIs with three priority tiers (HIGH/Medium/LOW).
- Implementation of FFmpeg-based adaptive compression with runtime codec switching supporting H.264, H.265, AV1, and NVENC.
- Risk-aware event engine with multi-signal risk score calculation and three-state surveillance machine (NORMAL, ALERT, CRITICAL).
- Pre/post-event circular frame buffer preserving up to 15 seconds of forensic context around critical incidents.
- Interactive live dashboard with canvas-based rendering, priority-coded ROI overlays, real-time sliders, preset modes, and live bandwidth charts.
- Central server with dual Socket.IO namespaces, UDP bandwidth monitoring, auto-bandwidth adaptation, and comprehensive CSV metrics logging.
- Automated evaluation pipeline computing global/per-ROI/background PSNR, SSIM, VMAF, detection accuracy, and investigation usability scores.
- Three privacy operation modes: background blur, ethical blanking, and face masking.
- Docker containerization with Docker Compose orchestration and YAML configuration profiles.
- Performance validation demonstrating 45-65% bandwidth reduction with stable quality metrics.

7.2 Outside Scope

- Multi-camera simultaneous deployment and scaling (architecturally supported but not validated for concurrent multi-node operation).
- Integration with commercial CCTV hardware or cloud platforms beyond standard IP camera interfaces.

- Mobile application development or voice-based alert systems.
- Production-grade security hardening including TLS/SSL encryption and penetration testing.
- Long-term field reliability studies or extended deployment trials.

8. Methodology

The proposed system follows a multi-stage adaptive video processing pipeline designed to reduce bandwidth consumption while preserving critical visual information. The architecture has evolved significantly from the initial concept to include risk-aware event-driven operation.

8.1 System Architecture

The system follows a three-tier architecture comprising the Edge Node (Python-based capture and processing pipeline), Central Server (Node.js with dual Socket.IO namespaces), and Dashboard (Next.js web application). The evaluation pipeline operates as a fourth component for automated benchmarking.



Fig 01: System Architecture — Three-tier edge-server-dashboard design with evaluation pipeline.

8.2 Frame Acquisition and YOLOv8 Object Detection

Video frames are captured from a live camera source (PC webcam or IP camera) using OpenCV. Each frame is processed in real time on the edge device. The YOLOv8n (nano) neural network performs object detection across 80 COCO classes, classifying detections into three priority tiers: HIGH (persons, class 0 — preserved at full resolution), MEDIUM (vehicles including cars, motorcycles, buses, trains, trucks — mildly compressed), and LOW (all other detected objects) [27]. Detection is performed at configurable intervals (default: every 3 frames) to balance accuracy and performance.

8.3 Temporal ROI Persistence and Anti-Flicker

A TTL (Time-to-Live) system ensures ROI bounding boxes persist for 30 frames (~2 seconds at 15 FPS), eliminating the pulsating flicker effect caused by periodic detection. A `merge_rois()` function using Intersection-over-Union (IoU) threshold ≥ 0.3 smoothly blends new detections into existing bounding boxes rather than hard-replacing them, providing temporally stable ROI boundaries.

8.4 Adaptive Compression and Encoding

Each frame is subdivided into foreground (ROI) and background (non-ROI) regions. ROI regions are encoded at high quality while the background is downscaled and aggressively compressed. The CodecManager class manages an FFmpeg subprocess that receives raw BGR frames via stdin and outputs an H.265 MPEG-TS stream over UDP. The codec can be switched at runtime between libx264, libx265, libsvtav1, and hevc_nvenc without restarting the subprocess. Adjustable parameters include Background Scale (resolution reduction), Background Quality (JPEG compression level), ROI Quality (high-quality encoding), and Frame Skipping (detection interval).

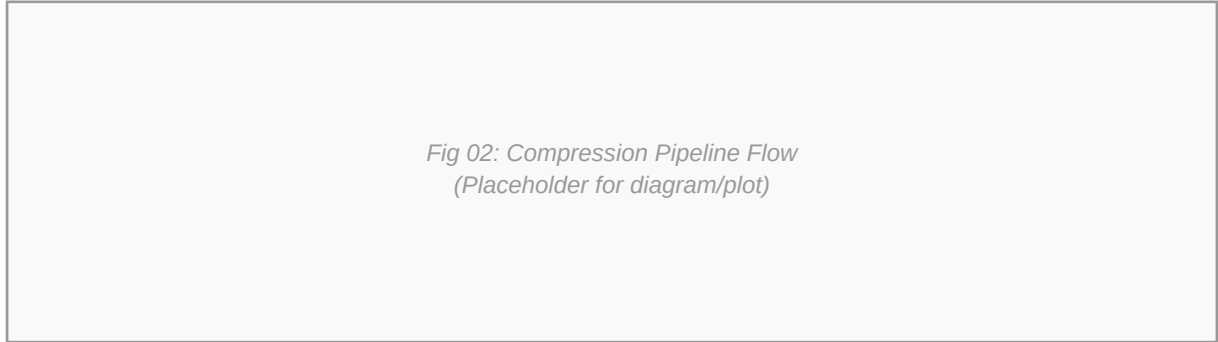


Fig 02: Adaptive compression pipeline showing ROI detection, background downscaling, and H.265 UDP streaming.

8.5 Risk-Aware Event Engine

The most significant architectural innovation is the risk-aware event engine. A `risk_score()` function computes a normalized risk value [0.0, 1.0] per frame by combining five independent signals: HIGH-priority objects (+0.30 each), MEDIUM-priority objects (+0.10 each), motion area fraction (up to +0.25), after-hours period 22:00-06:00 (+0.20 flat bonus), and scene change magnitude (up to +0.15). This score drives a three-state surveillance state machine:

State	Risk Range	Behavior	BG Scale	BG Quality	ROI Quality
NORMAL	0.00 - 0.25	Aggressive background compression; FPS halved if idle	User-defined	User-defined	User-defined
ALERT	0.25 - 0.55	Context-preserving; more background kept	0.75	45	95
CRITICAL	0.55 - 1.00	Near-lossless recording; full resolution	1.0	88	100

Hysteresis is implemented with an 8-frame hold period before downgrading to prevent rapid state oscillation. A rolling circular buffer (15 seconds x FPS) continuously stores original frames. When the state transitions to CRITICAL, pre-event frames are dumped to disk with full metadata, followed by 10 seconds of post-event recording, enabling forensic reconstruction of incident context.

8.6 Frame Encoding and Transmission

Background frame and ROI crops are encoded separately (Base64 JPEG). The data packet structure includes `frame_id`, `bg_data`, `rois` list with bounding boxes, `orig_w`, `orig_h`, `timestamp`, `risk_score`, and `surveillance_state`. Transmission uses Socket.IO for real-time streaming to the central server.

8.7 Server-Side Processing

The Node.js server manages two Socket.IO namespaces: /stream (edge node pushes frames) and /view (dashboard pulls frames and sends controls). A UDP server monitors H.265 stream bitrate in real time. Automatic bandwidth adaptation computes total bandwidth every 3 seconds and sends reduced-quality control messages if thresholds are exceeded. Three CSV logs maintain comprehensive records: metrics_log.csv (6000+ rows), control_log.csv (parameter changes), and motion_log.csv (40,000+ rows of per-frame motion data). HTTP endpoints include /control, /metrics, /sample, /reconstruct, /request_sample, /playback, /event, and /config.

8.8 Dashboard Visualization

Two visualization modes are implemented:

- **Live Monitor Mode:** Displays fully reconstructed frames on a dynamically-resized HTML5 canvas with priority-coded ROI overlay boxes (RED=HIGH, YELLOW=MEDIUM, GREEN=LOW) with shadowBlur glow effects.
- **Analysis Mode:** Displays compressed background and ROI overlays for analyzing compression behavior.

The dashboard features four real-time control sliders (BG Scale, BG Quality, ROI Quality, Detect Every N Frames), two preset mode buttons (Low BW and High Quality), dual bandwidth display (Raw Socket.IO kbps and H.265 UDP kbps with % Saved), and a live Chart.js bandwidth chart. A color-coded state badge (NORMAL green, ALERT orange, CRITICAL red) indicates the current surveillance state.



Fig 03: Live dashboard showing canvas stream, ROI overlays, control sliders, bandwidth chart, and state badge.

8.9 Metrics Collection and Analysis

The system continuously records and visualizes:

- FPS (Frames per second) — real-time frame rate display
- Bitrate (Kbps) — dual monitoring of Socket.IO raw stream and H.265 UDP stream
- Number of ROIs — count of detected objects per frame
- Risk Score — normalized [0.0, 1.0] surveillance risk indicator
- Surveillance State — NORMAL/ALERT/CRITICAL with hysteresis
- PSNR (Peak Signal-to-Noise Ratio) — global, per-ROI, and background-only
- SSIM (Structural Similarity Index) — global, per-ROI, and background-only
- VMAF Score — Netflix perceptual quality metric via FFmpeg libvmaf
- Detection Accuracy — YOLOv8 re-detection count retention on compressed frames

Fig 04: Performance Metrics Dashboard
(Placeholder for diagram/plot)

Fig 04: Real-time metrics visualization showing FPS, bitrate, ROI count, PSNR, SSIM, and risk score trends.

8.10 Compressed Frame Storage

Frames are stored in a structured directory format:

recording/ |-- [date]/ |-- bg/ (background images) |-- roi/ (ROI crops) |-- meta.json (metadata with bounding boxes and timestamps)

Background and ROI data are stored separately with metadata including bounding boxes and timestamps, enabling efficient storage and later reconstruction. The storage module also maintains event archives with pre/post-event frame sequences for forensic analysis.

Level	Component	Data Type	Description
Root	recordings/	Directory	Global container for recording sessions
L1	[date]/	Directory	Discrete session identifier (ISO 8601)
L2	/bg/	Binary/RAW	Background frame images
L2	/roi/	Binary/RAW	ROI crop images with bounding boxes
L2	meta.json	JSON	Associative metadata (timestamps, hardware IDs, risk scores)
Events	events/	Directory	Pre/post-event frame archives for critical incidents

8.11 Frame Reconstruction

Stored frames can be reconstructed by loading the background frame and overlaying ROI images at their corresponding bounding box coordinates. This ensures compression is selective rather than destructive, and stored surveillance footage remains fully usable for forensic review.

8.12 Performance Assessment

The automated evaluation pipeline iterates over configurable {bg_quality, roi_quality} presets, sends each as a live control command, and captures original/reconstructed frame pairs. For each experiment, global PSNR/SSIM, per-ROI PSNR/SSIM, background-only PSNR/SSIM, VMAF scores, and detection accuracy retention are computed. Visual artifacts including diff heatmaps and ROI overlays are saved. The event evaluation module computes investigation usability scores based on frame retention, detection recall, and pre-event context PSNR.

8.13 Methodology Overview

Phase	Functional Module	Technologies & Methods	Primary Deliverable
I	Edge Node	OpenCV, YOLOv8, FFmpeg, CodecManager	ROI-optimized compressed streams with risk-aware state machine
II	Central Server	Node.js, Socket.IO, UDP, HTTP, CSV logging	Real-time telemetry, auto-bandwidth adaptation, event logging
III	Dashboard	Next.js, React, Chart.js, HTML5 Canvas	Control UI, real-time visualization, state display
IV	Evaluation	scikit-image, FFmpeg libvmaf, YOLOv8	Automated benchmarking with comprehensive metrics
V	Deployment	Docker, Docker Compose, YAML configs	Reproducible containerized deployment

9. System Features

9.1 YOLOv8-Based Intelligent Object Detection

The system replaces conventional motion detection with YOLOv8n neural object detection across 80 COCO classes. Detected objects are classified into three priority tiers enabling semantically aware compression allocation. This provides significantly lower false-positive rates in complex lighting conditions compared to traditional frame-differencing approaches [27].

9.2 Real-Time Adaptive Compression with Multi-Codec Support

Runtime-switchable codec support enables selection between H.264, H.265, AV1, and NVENC encoders based on hardware availability and bandwidth requirements. Compression parameters adapt dynamically to both user input and automatic bandwidth detection.

9.3 Risk-Aware Surveillance State Machine

A multi-signal risk score combining object presence, motion area, time-of-day, scene changes, and ROI density drives a three-state machine with hysteresis. This ensures that during critical incidents, the system automatically transitions to near-lossless recording mode, preserving forensic evidence quality.

9.4 Pre/Post-Event Frame Buffer

A rolling circular buffer maintains 15 seconds of original full-resolution frames. When a critical event is detected, pre-event context and post-event recording are automatically archived with complete metadata, enabling forensic reconstruction of incident timelines.

9.5 Dual-Mode Dashboard Display

Live monitoring mode provides real-time observation with priority-coded ROI overlays and state badges. Analysis mode renders compressed background with ROI overlays for post-processing evaluation and research analysis.

9.6 Bandwidth-Aware Adaptive Control

The system dynamically adjusts compression parameters based on network bandwidth measurements. The server automatically reduces quality when bandwidth thresholds are exceeded, while user-initiated controls take priority and suppress auto-adaptation for a configurable period.

9.7 Temporal ROI Persistence

TTL-based ROI persistence with IoU-smoothing merge algorithm eliminates visual flickering caused by periodic detection. ROIs persist for 30 frames (~2 seconds) with smooth transitions between detection cycles.

9.8 Privacy and Ethical Operation Modes

Three privacy tiers address ethical surveillance concerns: Privacy Blur (heavy Gaussian background blur), Ethical Mode (black frame when no ROI detected — zero passive surveillance), and Face Masking (anonymization of detected persons while preserving vehicle visibility).

9.9 Comprehensive Evaluation Pipeline

The evaluation system computes signal quality (PSNR/SSIM), perceptual quality (VMAF), detection accuracy retention, and investigation usability scores. Per-ROI and background-only metrics provide granular insight into compression behavior.

9.10 Docker Containerization and Configuration Profiles

Full Docker containerization with Docker Compose orchestration enables single-command deployment. Four YAML configuration profiles (balanced, high_quality, privacy_mode, ultra_low_bandwidth) provide pre-configured operation modes for different deployment scenarios.

9.11 Functional Requirements

ID	Category	Requirement	Technical Description
FR-01	Ingestion	Video Stream Input	Capture live frames from webcam or IP camera using OpenCV
FR-02	Analysis	YOLOv8 Object Detection	Detect and classify objects across 80 COCO classes with three priority tiers
FR-03	Analysis	Risk Score Computation	Calculate normalized risk score from object presence, motion, time, scene change, and density signals
FR-04	Processing	ROI-Based Adaptive Compression	Segment frames into ROI and background with per-region quality allocation
FR-05	Processing	State Machine Management	Three-state surveillance machine with hysteresis (NORMAL/ALERT/CRITICAL)
FR-06	Processing	Pre/Post-Event Buffer	15-second rolling buffer with automatic event-triggered archival
FR-07	Encoding	Multi-Codec Encoding	Runtime-switchable codec: H.264, H.265, AV1, NVENC via FFmpeg
FR-08	Networking	Real-Time Transmission	Dual Socket.IO namespaces + H.265 UDP streaming
FR-09	Control	Dynamic Parameter Adjustment	Real-time modulation of BG Scale, BG Quality, ROI Quality, detection intervals
FR-10	Control	Auto Bandwidth Adaptation	Automatic quality reduction when bandwidth thresholds exceeded

ID	Category	Requirement	Technical Description
FR-11	Visualization	Live Monitoring	Canvas-based stream rendering with priority-coded ROI overlay and state badge
FR-12	Visualization	Analysis Interface	Secondary mode for compressed background with ROI overlays
FR-13	Analytics	Performance Metrics	FPS, bitrate, ROI count, PSNR, SSIM, VMAF, detection accuracy
FR-14	Storage	Frame Archival	Structured storage with separate BG/ROI/meta and event archives
FR-15	Reconstruction	Video Synthesis	Composite reconstruction from background and ROI segments
FR-16	Privacy	Privacy Operation Modes	Background blur, ethical blanking, face masking

9.12 Non-Functional Requirements

Dimension	Specification	Success Criteria / Metrics
Performance	Real-Time Processing	End-to-end latency 200-500 ms; dashboard initial UI load ~3s; sub-second telemetry refresh
Reliability	Continuous Operation	Sustained uptime without degradation; automatic socket reconnection; packet loss resilience
Usability	Interface Design	Cohesive UI/UX with standardized layouts across monitoring, control, and analytical views
Maintenance	Modular Architecture	Decoupled 3-layer design (Edge, Server, Dashboard) ensuring independent extensibility
Scalability	Multi-Stream Support	Architectural framework validated for single-stream with structural support for multi-node scaling
Efficiency	Bandwidth Optimization	Targeted data reduction of 45-65% relative to unoptimized full-frame transmission
Observability	Logging & Monitoring	Automated archival of system events and performance metrics in standardized CSV formats
Security	Access Perimeter	Deployment within trusted LAN with roadmap for future TLS/SSL integration
Privacy	Data Protection	No facial recognition; no persistent raw footage storage; clear ethical operation modes

10. Project Timeline

Sl.	Activity	Duration	Start Date	End Date	Status
1	Idea Generation, Requirement Analysis and Feasibility Study	2 Weeks	20 Oct 2025	02 Nov 2025	Completed
2	Literature Review and Research Planning	3 Weeks	03 Nov 2025	23 Nov 2025	Completed
3	System Design (Architecture and ROI Pipeline Design)	2 Weeks	24 Nov 2025	03 Dec 2025	Completed

Sl.	Activity	Duration	Start Date	End Date	Status
4	Prototype Development (Edge Setup, Basic Dashboard)	10 Days	28 Nov 2025	07 Dec 2025	Completed
5	Poster Presentation (First Prize Achieved)	3 Days	05 Dec 2025	08 Dec 2025	Achieved
6	Edge Node Refinement (YOLOv8 Integration, FFmpeg)	3 Weeks	09 Dec 2025	29 Dec 2025	Completed
7	Server-side Development and Advanced Metrics Logging	3 Weeks	30 Dec 2025	19 Jan 2026	Completed
8	Dashboard Enhancement (Graphs, Controls, UI/UX)	3 Weeks	20 Jan 2026	09 Feb 2026	Completed
9	System Integration (Edge, Server, Dashboard)	3 Weeks	10 Feb 2026	02 Mar 2026	Completed
10	Real-Time Testing and Performance Evaluation	3 Weeks	03 Mar 2026	23 Mar 2026	Completed
11	Risk-Aware Event Engine Development	2 Weeks	24 Mar 2026	06 Apr 2026	Completed
12	Docker Containerization and Deployment Automation	2 Weeks	07 Apr 2026	20 Apr 2026	Completed
13	Documentation, Thesis Writing and Final Reporting	4 Weeks	21 Apr 2026	18 May 2026	In Progress

*Fig 05: Gantt Chart — Project Timeline
(Placeholder for diagram/plot)*

Fig 05: Gantt chart illustrating project schedule from October 2025 to May 2026.

11. Ethical Consideration

Ethical considerations ensure that the technologies employed in this project — particularly CCTV surveillance video processing, YOLOv8-based object detection, and AI-driven adaptive compression — are used in a safe, responsible, and legally compliant manner. The system processes live video streams and does not perform facial recognition, individual identification, or any form of biometric analysis. The system detects generic objects to enable dynamic ROI compression and risk-aware state transitions.

11.1 Privacy Protection in CCTV Data Processing

The system uses CCTV video streams exclusively for real-time adaptive compression and risk-aware event detection. No personally identifiable information, identity data, or sensitive content is stored beyond what is minimally required for motion detection and compression. All video processing occurs at the edge node, and only authorized personnel can access live streams

through the dashboard's role-based access control.

11.2 No Facial Recognition or Individual Identification

The system performs no facial recognition or individual identification. YOLOv8 detection classifies objects at the category level (person, vehicle, etc.) without attempting to identify specific individuals. This ensures the system does not infringe on personal privacy or enable unauthorized surveillance monitoring of individuals.

11.3 Transparent Operation and Algorithmic Accountability

The YOLOv8 object detection logic, risk score computation, and FFmpeg compression pipeline are completely transparent and auditable. The multi-signal risk score is explainable (each component contribution is logged), and the state machine transitions are deterministic. False detections and their impact on compression behavior can be traced and analyzed.

11.4 Privacy-First Operation Modes

Three privacy modes are built into the system by design: Privacy Blur applies heavy Gaussian blur to the entire background preventing any background surveillance; Ethical Mode returns a black frame when no ROI is detected, ensuring zero passive surveillance of empty spaces; Face Masking anonymizes detected persons while preserving vehicle visibility for security monitoring.

11.5 Secure Storage and Data Retention

Only compressed temporary frames are stored. No continuous raw video recording occurs unless triggered by the event engine during critical incidents. Access to compressed streams, event archives, and metrics logs is restricted to authorized personnel through the dashboard's authentication layer.

11.6 Ethical Use of Test Data

All test footage used during development was captured in controlled laboratory settings or sourced from publicly available, non-sensitive datasets with appropriate permissions. No personally identifiable information or private surveillance data from third parties was utilized. The YOLOv8 model was used with pre-trained COCO weights, and no additional training on sensitive surveillance data was performed.

12. Innovation

Compared to conventional CCTV surveillance systems and fixed-bitrate compression methods, this Adaptive CCTV Compression System using Dynamic ROI Control introduces several novel contributions:

12.1 Risk-Aware Event-Driven Compression

The most significant innovation is the transformation from a generic adaptive compressor into a mission-critical, event-triggered surveillance system. The multi-signal risk score combining object presence, motion area, time-of-day, scene changes, and ROI density produces a principled, explainable risk metric. The three-state machine with hysteresis ensures appropriate compression behavior for each surveillance context — aggressive compression during idle periods, context

preservation during suspicious activity, and near-lossless recording during critical incidents.

12.2 Pre/Post-Event Forensic Frame Buffer

The rolling circular buffer preserving 15 seconds of pre-event and 10 seconds of post-event full-resolution frames addresses a critical gap in existing surveillance compression systems. Investigators typically need the moments before an incident to understand how events unfolded. No existing ROI-based compression system in the reviewed literature provides this capability.

12.3 YOLOv8-Based Priority-Aware Compression

Moving beyond basic motion detection, the system uses YOLOv8 neural object detection with three-tier priority classification. Persons are preserved at full resolution, vehicles receive moderate compression, and other objects undergo aggressive compression. This semantic awareness ensures that the most important surveillance elements receive the highest quality allocation.

12.4 Temporally Stable ROI with IoU Smoothing

The TTL-based ROI persistence with IoU merge algorithm eliminates the visual flickering that plagues periodic detection systems. Smooth transitions between detection cycles provide a stable viewing experience while maintaining compression efficiency.

12.5 Comprehensive Investigation Usability Metric

The project introduces a novel composite investigation usability score that measures whether compressed surveillance footage would be usable as forensic evidence. Combining frame retention rate, detection recall, and pre-event context quality, this metric answers the question that truly matters for surveillance: not just compression ratios, but evidence preservation utility.

12.6 Runtime-Switchable Multi-Codec Architecture

The CodecManager enables runtime codec switching without subprocess restart, supporting H.264, H.265, AV1, and NVENC. This flexibility allows the system to adapt to different hardware capabilities and deployment requirements without service interruption.

12.7 Integrated Privacy-by-Design

Rather than treating privacy as an afterthought, the system incorporates three distinct privacy modes built directly into the compression pipeline. This ethical-by-design approach ensures that deployment in sensitive environments can be configured for appropriate privacy protection without sacrificing security monitoring capabilities.

13. Expected Outcomes

- A fully functional real-time adaptive CCTV compression system with YOLOv8-based dynamic Region of Interest (ROI) control and risk-aware event-driven state machine.
- Bandwidth reduction of 45-65% achieved with stable PSNR (30-35 dB), SSIM (0.85-0.95), and VMAF scores verified through automated evaluation pipeline.
- Working prototype deployed on an edge node with live video feed, demonstrated through interactive web dashboard with real-time control and visualization.

- Pre/post-event forensic frame buffer enabling evidence preservation during critical security incidents with investigation usability scoring.
- User-controlled live dashboard with real-time monitoring of bitrate, PSNR, SSIM, VMAF, risk score, surveillance state, and compression parameters.
- Comprehensive evaluation dataset with 33,880+ experiment data points, 6000+ metrics log entries, and 40,000+ motion log entries.
- Docker containerized full-stack deployment with four YAML configuration profiles for different operational scenarios.
- Full source code, comprehensive documentation, test datasets, and performance evaluation reports.
- Research poster and technical paper for publication in academic venues.

14. Estimated Budget

Item	Quantity	Price (Taka)
IP Camera (1080p)	1	3,100
Edge Device (Raspberry Pi / Laptop)	1	0 (Available)
Server Setup (Local Machine)	1	0 (Available)
SSD (500GB for storage and recording)	1	2,000
Dashboard Development	-	0 (In-house)
YOLOv8 Model License	-	0 (Open source)
Software Libraries (OpenCV, FFmpeg, etc.)	-	0 (Open source)
Networking Equipment (Router, Cables)	1 set	1,500
Miscellaneous (Power, connectors, etc.)	-	1,000

15. Conclusion

This project presents a practical and innovative solution to one of the most pressing challenges in modern surveillance systems: the inefficiency of fixed-bitrate video streaming and the lack of intelligent, event-aware compression. The Adaptive CCTV Compression System with Dynamic Region of Interest (ROI) Control successfully integrates YOLOv8-based object detection, FFmpeg adaptive encoding with multi-codec support, a risk-aware surveillance state machine, and an interactive real-time dashboard to achieve significant bandwidth savings while preserving visual quality in critical areas and during important events.

The implemented system has evolved substantially beyond the originally proposed scope. Key enhancements include the replacement of basic OpenCV motion detection with YOLOv8 neural object detection with priority classification, the addition of temporal ROI persistence eliminating visual flicker, the implementation of a multi-signal risk score and three-state surveillance machine, the pre/post-event forensic frame buffer for evidence preservation, three privacy operation modes, runtime-switchable codec support, Docker containerization, and a comprehensive evaluation pipeline with VMAF scoring, detection accuracy retention, and investigation usability metrics.

The system is strongly suited for deployment in smart cities, educational institutions, industrial facilities, and low-resource IoT environments in Bangladesh. The edge-based architecture, stable performance metrics, privacy-preserving design, and user-friendly dashboard make direct contributions to the Digital Bangladesh and Smart Bangladesh Vision 2041 initiatives. The project's novel contributions — particularly the risk-aware event engine and the investigation usability metric — represent meaningful advances beyond the current state of the art in surveillance video compression.

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AI Check Ratio

This proposal has been prepared through rigorous research, analysis, and implementation work conducted by the project team. AI-assisted tools may have been used for literature review organization and editing assistance. The technical content, implementation, experimental results, and conclusions represent original work by the project team members.

AI Similarity Check Report (Placeholder)
(Placeholder for diagram/plot)

AI-generated content similarity analysis report will be attached upon final submission.